

David Meredith

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EDUCATION

University of Utah, B.S. Computer Science

May 2026

Minor in Mathematics

- **GPA:** 3.991, *Magna Cum Laude*
- **Relevant Courses:** Codes and Cryptography, Practical Machine Learning, Software Development, Mobile Application Programming, Web Dev I, Database Systems, Number Theory, Algorithms, Data Structures, Foundations of Analysis

PROJECTS

Nocturne: Escape the Night (CS Capstone Project)

August 2025 – April 2026

C++, Unreal Engine

- Jointly developed a year-long project in an Agile 15-person interdisciplinary team of engineers, artists, and designers.
- Guided design and architecture discussions for the project by leveraging prior experience with the software to advocate for event-driven communication using Unreal Engine delegates to improve system organization and maintainability.
- Designed 5 core C++ system, independently implementing 3 and worked with another engineer in the other 2 to achieve a working prototype in early game development stages.
- Debugged multilayered function calls to resolve complex pointer and memory issues avoiding repeated game crashes.
- Refactored existing code helping remove redundancies and improve readability by consolidating repeated logic into reusable structures for team collaboration and smoother future adjustments.

Drawing App (Final Project)

Kotlin, Android Studio, Firebase

September 2024 – December 2024

- Integrated Firebase backend services to enable image upload, storage, and real-time sharing between users for the app.
- Managed user authentication and image management backend systems through iteration to allow early stage app usage.
- Coordinated project tasks and led troubleshooting efforts to ensure completion by the project deadline.

Image Classification System (Machine Learning Final Project)

Python, PyTorch

March 2026 – April 2026

- Implemented a lightweight CNN (LeNet-5) in PyTorch to perform image classification on the CIFAR-100 dataset.
- Implemented training and validation pipelines, including batching and dataset handling for efficient model training.
- Developed analytical tools for machine learning networks to analyze model performance and justified results using visualizations (line graphs, heatmaps) to track loss and accuracy across hyperparameter configurations.

WORK EXPERIENCE

Teaching Assistant (Software Practice II)

August 2024 – April 2026

University of Utah | Salt Lake City, UT

- Instructed lab sessions for 20+ students per session, covering C++ programming and Linux environments in Docker.
- Evaluated 50+ students' assignments and projects weekly, delivering targeted feedback on code quality to strengthen skill development.
- Guided students through programming concepts and project debugging during office hours, reinforcing systematic troubleshooting practices for stronger code quality.
- Collaborated with faculty to improve course materials, ensuring clear documentation and alignment with software design best practices for a stronger student learning experience.

Administrative and Operations Intern

November 2023 – May 2024

Ken Garff Success in Education | Salt Lake City, UT

- Compiled and delivered monthly data reports to provide clear insights and support data-driven planning.
- Interfaced with elementary school administrations to coordinate information and ensure accurate program records.

SKILLS

Languages – C++, C#, Python, C, SQL, HTML, Kotlin, Java

Infrastructure & Tools – Visual Studio Code, Docker, Firebase, MySQL, Qt Creator, Linux, .NET, Swift, Claude Code

Data – NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-learn

Version Control – Git, GitHub, Perforce